

NDFEX Bot Reference

Exchange Bots & Market Simulation

This document describes the automated bots that run on the NDFEX exchange during the trading competition. These bots provide liquidity, generate price movement, and simulate realistic market conditions for teams to trade against.

1. Overview

Two bot processes run concurrently on the exchange, each logged in as a separate user:

Process	User	Role
bot_runner	test (ID 99)	Primary market maker — provides liquidity and drives price discovery via random-walk fair values
smarter_bots	test2 (ID 98)	Reactive trader — responds to market conditions with momentum, pressure, and passive order flow

Both processes trade all 13 symbols. Together they create a market with continuous two-sided quotes, random price movement, and periodic aggressive order flow.

2. bot_runner Bots

The bot_runner process contains three bot types that work together to form the core market structure.

2.1 RandomWalkFairValue

Not a bot itself, but the price engine that drives the market makers. Each of the 12 standalone symbols (GOLD, BLUE, and the 10 dorms) has its own independent random walk that evolves over time.

How it works:

- Starts at a configured initial price (e.g. GOLD at 1200, BLUE at 900, dorms in the 500–600 range).
- At random intervals (1–100ms), the fair value moves up or down by a small random step (bounded by tick size).
- The value is clamped to stay within the symbol's min/max price range.

The UNDY ETF uses a separate **BasketFairValue** that computes its fair value as the sum of the 10 dorm component mid-prices from the live order book, just like a real ETF arbitrageur would.

2.2 FairValueMarketMaker

The primary liquidity provider. Seven instances run simultaneously, each with a different "variance" (bias) applied to the fair value, creating a spread of opinions in the market.

How it works:

- Computes a biased fair value: $fv = random_walk_fv + variance$.
- Applies position-based skew: the fair value is adjusted downward by the bot's current position (up to ± 100), so large positions push quotes away to reduce inventory.
- Places a resting bid and ask a configurable number of ticks wide of the skewed fair value.
- Cancels and replaces orders when the fair value moves or after 30 seconds, whichever comes first.

The seven market maker personalities:

Instance	Variance Profile	Width	Qty
1	Neutral (all zeros)	2 ticks	100
2	Slightly bullish (+1 all symbols)	2 ticks	90
3	Slightly bearish (-1 all symbols)	2 ticks	80
4	Mixed (+2/0/+1 alternating)	3 ticks	70
5	Alternating (+2/0 pattern)	3 ticks	60
6	Dorm focused (+2 dorms, 0 GOLD)	3 ticks	50
7	Contrarian (+3 GOLD, ± 1 dorms)	3 ticks	40

All UNDY variances are 0 — the ETF is always quoted at basket fair value without bias.

2.3 FairValueStackingMarketMaker

A depth-of-book liquidity provider that adds resting orders at multiple price levels behind the best bid/ask. One instance runs per symbol (13 total).

How it works:

- Computes the **weighted mid price** from the current best bid/ask and their quantities.
- Uses a **StackManager** to maintain a stack of up to 10 resting orders on each side, spaced 1 tick apart.
- Orders are placed at 1 tick wide of the weighted mid, with 2 lots each.
- As the mid moves, the stack re-centers: stale levels are cancelled and new levels are added.

This creates realistic depth in the order book beyond the top of book.

2.4 RandomTaker

Simulates uninformed aggressive order flow. Four instances run with different quantities and timing scales.

How it works:

- At random intervals, sends **IOC** (Immediate or Cancel) orders that cross the spread to trade immediately.
- Each tick, randomly chooses buy or sell for every symbol.
- Buys at the best ask price; sells at the best bid price.

Instance	Quantity	Interval Range
1	1 lot	60–3,000 ms
2	2 lots	70–3,500 ms
3	3 lots	80–4,000 ms
4	4 lots	90–4,500 ms

RandomTakers generate volume and trigger fills against the market makers' resting orders, creating trade flow that moves prices.

3. smarter_bots

The smarter_bots process runs three reactive bot types that respond to live market conditions rather than following a random walk. These bots create momentum, directional pressure, and passive order flow.

3.1 ImbalanceTaker

A momentum-following bot that detects order flow imbalance and trades in the direction of the dominant side.

How it works:

- Listens to the trade summary feed and tracks a running imbalance counter per symbol: buy-side volume increments the counter, sell-side decrements it.
- When the accumulated imbalance exceeds a per-symbol threshold, the bot sends an aggressive order in the direction of the imbalance (buy if imbalance is positive, sell if negative).
- After firing, the imbalance counter for that symbol resets to zero.
- A 100ms cooldown prevents rapid-fire orders.

Parameter	Value
Order quantity	40 lots per trigger
Cooldown	100 ms between orders
Order type	Aggressive (max/min price to ensure fill)

Imbalance thresholds by symbol:

Symbol	GOL D	BLU E	KNA N	STE D	FISH	DILN	SOR N	RYA N	LYO N	WLS H	LEWI	BDIN	UND Y
Thresho ld	200	400	300	300	300	300	300	300	300	300	300	300	250

GOLD has the lowest threshold (200) making it most responsive to imbalance. BLUE has the highest (400), requiring larger one-sided flow before the bot reacts.

3.2 PressureTaker

Simulates directional pressure by sending a burst of aggressive orders on one side of the market, pushing prices in a single direction.

How it works:

- Periodically loads a random "pressure campaign": a random quantity (1–100 lots) and a random direction (buy or sell).
- Sends **IOC** orders of 5 lots across all 13 symbols on the chosen side, repeatedly, until the pressure quantity is exhausted.
- Each burst is separated by a random delay of 200ms–10 seconds.
- Buys hit the best ask; sells hit the best bid. If no liquidity exists on the target side, that symbol is skipped.

The effect is a sustained push in one direction across all symbols simultaneously, followed by a quiet period, then a new random direction. This creates momentum-style moves that teams can detect and trade around.

3.3 RestingOrders

Simulates a passive participant that places limit orders away from the current market, adding depth to the book.

How it works:

- At random intervals (500ms–10 seconds), picks a random side (buy or sell).
- Places resting orders on that side across all 13 symbols.
- Buy orders are placed **3 ticks below** the current best bid; sell orders are placed **3 ticks above** the current best ask.
- Each order is for 25 lots.
- Orders are fire-and-forget — they are never cancelled by this bot, so they accumulate in the book as passive liquidity that can be traded against.

These resting orders create opportunities for teams: they sit away from the current price and can be picked off if the market moves toward them.

4. How the Bots Interact

The two bot processes create a self-sustaining market ecosystem:

- **FairValueMarketMaker** provides continuous two-sided quotes that track the random walk, establishing the "fair" price for each symbol.
- **FairValueStackingMarketMaker** adds depth behind the top-of-book, making the order book realistic.
- **RandomTaker** generates baseline trade flow by crossing the spread, causing fills and price prints.
- **ImbalanceTaker** amplifies momentum: when one-sided flow accumulates, it piles on, creating short-term trends.
- **PressureTaker** creates sustained directional moves across all symbols at once.

- **RestingOrders** seeds the book with passive liquidity away from the market, creating depth and fill opportunities for teams.
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5. What This Means for Teams

- **There is always liquidity.** The market makers ensure continuous two-sided quotes in all symbols.
- **Prices move.** The random walk, taker bots, and pressure bursts create genuine price action.
- **Momentum is real.** The ImbalanceTaker and PressureTaker create short-term trends you can detect.
- **Depth exists away from the market.** RestingOrders place passive orders 3 ticks out that can be filled if the market moves.
- **The ETF tracks its components.** UNDY quotes are derived from live dorm mid-prices, so ETF arbitrage opportunities arise when dorm prices move.
- **Spreads vary.** The seven market makers have different widths and biases, so the effective spread varies by symbol and over time.